Supplementary Information: Genome-scale context, stoichiometric refinement, and neuro-symbolic reasoning for proteome function annotation

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S1 Benchmark suite overview, Stage 0 calibration, and Suite 1 evaluations

Supplementary Table S1 summarizes the eight benchmark suites evaluated in the main manuscript, including their cohorts, Stage 0 evidence channel configurations, active pipeline stages, primary comparators, and canonical result tables in the repository. Supplementary Table S2 reports the ancestor-propagated evaluation across the 13-genome and 21-genome reference panels comparing GSPA against production annotation pipelines, individual Stage 0 channels, and genomic language models.

Table S1. Overview of the eight GSPA evaluation suites, cohorts, Stage 0 channel configurations, active pipeline stages, and canonical result tables.

Suite	Evaluation Scenario & Cohort	Proteomes / Proteins	Stage 0 Channels	Active Stages	Primary Comparators	Canonical Result Table(s)
Suite 1A	CAFA3→CAFA5 Temporal GO Holdout	16,125 evals (3,440 at < 30% ID)	5-Ch (DIAMOND, Pfam, InterPro, ESM-2, ProtT5)	0 + 1 + 2A	DIAMOND, ProteInfer, DeepGOCNN, DeepGOPlus, DeepGOZero	scenario1_cafa_temporal_holdout_real.tsv
Suite 1B	Prospective CAFA6 Sequence Holdout	769 prospective targets	7-Ch (ESM-2, ProstT5 3Di, DIAMOND, InterPro, DeepGO-SE)	0 + 2A + 3	ESM-2+3Di MLP, Qwen3-14B Agent	suite1_cafa6_prospective_summary.tsv
Suite 1C	Post-Cutoff Swiss-Prot EC (Holdout-125)	125 enzymes (74 species, 293 pairs)	1-Ch or 3-Ch (CLEAN, DeepProZyme, EnzBERT)	0 + 1 + 2B	CLEAN, DeepProZyme, EnzBERT, METEOR v8	scenario1_swissprot_ec_holdout125_real.tsv
Suite 1D	95-Genome Bacterial EC Homology Masking	95 genomes (69,666 proteins; 4 ID tiers)	1-Ch or 3-Ch (CLEAN, DeepProZyme, EnzBERT)	0 + 1 + 2B	Standalone Predictors, METEOR v8	scenario1_95genomes_ec_homology_real.tsv
Suite 2A	9-Genome Stage 1/2A Ablation & Perturbation	9 genomes (81 runs; 900 injected claims)	3-Ch (DIAMOND, HMMER/Pfam, InterProScan)	0 + 1 + 2A	Stage 0 Noisy-OR, Individual Priors B_1–B_6	pillar1_2_ablation_summary.tsv
Suite 2B	13/21-Genome SOTA & 88-Proteome Multi-Kingdom	13, 21, and 88 proteomes (4 domains)	3-Ch or 7-Ch (Full Neural + Structural)	0 + 1 + 2A	NCBI PGAP, InterProScan, deepFRI, gLM, gLM2	pillar1_suite2_panel13_21_sota.tsv; multikingdom_summary_by_domain.tsv
Suite 2C	1,000-Genome Bacterial Population Audit	1,000 RefSeq bacterial genomes	3-Ch (DIAMOND, HMMER/Pfam, InterProScan)	0 + 1 + 2A	DeepGO-SE, SPROF-GO, DeepGraphGO, DeepFRI, TALE, InterProScan	pillar1_suite2_1000bacteria_sota.tsv
Suite 2D	6-Eukaryote Anatomical Bridge & Relaxation	6 species (6,155 endog.; 7,655 stress)	7-Ch (Full Neural + Structural)	0 + 1 + 2A	Direct GO Taxon Rules, Stage 2A Term Deletion	pillar3_eukaryote_anatomy.tsv; pillar3_relax_vs_remove.tsv
Suites 3–5	Role-Stratified LRO & Heteromeric Complex LSO	256/414 (*E. coli*), 144 (*B. subtilis*), 175 LSO	Track A (No PLM) or Track B (+ ESM-2 Centroids)	0 + 1 + 2B + 3	Sequence-Only/METEOR, RLGC Alone, GBDT	pillar2_lro15_summary.tsv; pillar3_complex_lso.tsv
Suite 6	RB-TnSeq Fitness Enzymes (Price-149) & 108-Audit	148 proteins (22 hosts); 108 GEMs; 6 BiGG	1-Ch or 5-Ch (CLEAN, MAPred, DPZ, EnzBERT, GraphEC, TopEC)	0 + 1 + 2B	6 Standalone Tools, METEOR v8, DCGM, ECtocr	pillar2_price149_all6_real.tsv; pillar2_invariant_audit_summary.tsv
Suites 7–8	BacDive Carbon Growth & Diagnostic Enzymes	529 (2,332), 3,017 (17,496), 6,313 (50,075)	3-Model Pool (gapseq, ModelSEED, CarveMe)	0 + 1 + 2B	CarveMe, ModelSEED, gapseq, METEOR, Traitar, DiASPora	pillar3_bacdive_track1_summary.tsv; pillar3_bacdive_track2_summary.tsv
Stage 3	De-Identified 120-Anomaly & 50-LRO Benchmark	120 anomalies (7 taxa); 50 LRO gaps (3 seeds)	Post-Stage-2 Candidate Set	1 + 2 + 3	Threshold ≥ 0.35, 5-Fold CV LR/RF/GBDT/MLP, No-Guardrail	pillar4_agent_evaluation.tsv; pillar4_multiseed_summary.tsv

Table S2. Ancestor-propagated evaluation across the 13-genome prokaryotic panel (11 prokaryotes), 21-genome multi-kingdom panel, and 9-genome genomic language model subset.

Panel Cohort	Method / Channel	Genomes (n)	Micro $F_{\max}$	CAFA $F_{\max}$	$\mathbf{WF}_{\max}$	$\mathbf{S}_{\min}$	EC $F_{\max}$	Runtime / Peak RAM
13-Genome Panel (Prok. 11)	metagenomic-deepFRI (2026)	11	0.6010	0.5940	0.1565	95.20	n/a	38.0 s / 4,800 MB
	InterProScan 5.72	11	0.7410	0.6780	0.4015	58.40	n/a	680.0 s / 8,200 MB
	NCBI PGAP (2021)	11	0.7820	0.7680	0.4438	54.20	n/a	1,800.0 s / 16,000 MB
	GSPA Stage 1+2 (3-Channel Full)	11	0.8421	0.8605	0.7215	21.40	0.8830	5.4 s / 4,812 MB
21-Genome Panel (All 21)	FoldSeek + ProstT5 3Di (2024)	21	0.6840	0.6310	n/a	83.24	n/a	10,800.0 s / 14,000 MB
	InterProScan 5.72	21	0.7180	0.6490	n/a	95.93	0.3120	720.0 s / 8,200 MB
	ESM2-DeepGOPlus (2024)	21	0.7420	0.6950	n/a	105.29	n/a	420.0 s / 6,400 MB
	ProteInfer (2023)	21	0.7950	0.7540	n/a	38.12	0.3880	120.0 s / 5,800 MB
	DIAMOND-BLASTp	21	0.8400	0.7890	n/a	93.81	0.8550	17.0 s / 2,100 MB
	ProtT5 k-NN (2021)	21	0.8530	0.8010	n/a	39.40	n/a	28.0 s / 6,200 MB
	ESM-2 k-NN (2023)	21	0.8580	0.8080	n/a	37.85	n/a	24.0 s / 5,800 MB
	GSPA Stage 1+2 (3-Channel Core)	21	0.8690	0.8220	n/a	36.14	0.8830	6.1 s / 4,812 MB
	GSPA Stage 1+2 (7-Channel All-Pooled)	21	0.8705	0.8407	n/a	35.20	0.8860	6.8 s / 4,812 MB
9-Genome Subset	gLM / gLM2 Genomic LM (2024)	9	0.7280	0.7610	n/a	37.40	n/a	95.0 s / 7,200 MB

S1.1 Stage 0 Platt calibration parameters and correlation groups

In Stage 0 (`ScoreCalibrator.groovy` and `LikelihoodModel.groovy`), raw predictor scores $s(c) \in [0, 1]$ are mapped via Platt scaling $\tilde{q}(c) = \text{clip}(\sigma_{\log}(a_e s(c) + b_e), 10^{-4}, 0.9999)$ and multiplied by channel reliability $r_e \in (0, 1]$. Default calibration slopes and intercepts (a_e, b_e) are $(7.0, -3.5)$ for sequence alignment (DIAMOND/BLAST), $(6.5, -2.9)$ for profile HMMs (HMMER/Pfam), $(6.0, -2.4)$ for InterProScan and KofamScan, $(6.8, -3.2)$ for ESM-2 and ProtT5 k-NN transfer, $(6.2, -2.8)$ for ProteInfer, DeepGOWeb, and DeepGO-SE, and $(6.4, -3.0)$ for FoldSeek and ProstT5 3Di structural transfer. Default channel reliability weights r_e (`EvidenceType.groovy`) are 0.85 for AMR domain predictors, 0.80 for sequence homology and orthology, 0.75 for profile HMMs and sequence domains, 0.70 for protein-sequence ML, PLM k-NN, and structural similarity, 0.65 for structure-based deep learning, 0.60 for metabolic pathway context and inferred context, 0.55 for genomic language models, and 0.45 for raw genomic neighborhood context. Claims are partitioned into 11 correlation groups in `LikelihoodModel.groovy`: homology, protein-sequence ML, structure, genomic context, genomic language models, inferred context, sequence motifs, localization, region, viral, and specialized domain predictors.

S2 Hyperparameter specification and Stage 1–2A implementation crosswalk

All 18 configurable pipeline hyperparameters across Stage 1 (13 configuration keys exposed by `IntegrateCommand.groovy` and consumed by `IterativeRefiner.groovy` and the Stage 1 prior modules, alongside the fixed candidate partner gate 0.08 in `CoherencePrior.groovy`, the fixed co-operonic complex partner bonus 0.95 in `GenomicContextPrior.groovy`, the Stage 2A ancestor relaxation factor 1.0 in `ConsistencyEnforcer.groovy`, and the Stage 2A obligate-complex singleton prune threshold 0.45 in `IntegrateCommand.groovy`) and Stage 2B (5 parameters in `stoichiometric_locus_refiner.py`, alongside the fixed elemental-imbalance penalty $w_{\text{imb}} = 100.0$) were calibrated on the *Escherichia coli* K-12 MG1655 development proteome and held frozen across all out-of-sample prokaryotic, archaeal, viral, and eukaryotic benchmarks. Supplementary Table S3 reports these 18 configurable hyperparameters, their configuration keys, source modules, roles in inference, and default values (where penalty parameters are stored as positive magnitudes and applied with negative sign in log-odds space).

S2.1 Temporal GO and EC holdouts and 95-genome homology masking

Supplementary Table S4 reports the CAFA temporal holdout benchmark across 16,125 protein–aspect evaluations (3,440 at $< 30\%$ sequence identity). Supplementary Table S5 reports the 16-row post-cutoff Swiss-Prot enzyme benchmark (Holdout-125, $n = 125$ proteins across 293 ground-truth protein–EC pairs), separating standalone per-protein baselines, single-predictor refinement, and three-method ensemble pooling. Supplementary Table S6 reports the 12-row enzyme evaluation across the 95-genome bacterial panel under four donor-database homology-masking tiers (unfiltered, $\leq 70\%$, $\leq 50\%$, and $\leq 30\%$ sequence identity), comparing unfiltered METEOR v8 Top-1 changes against GSPA's single-predictor stoichiometric and pathway acceptance filter and GSPA's three-channel consensus acceptance filter, alongside exact discordant protein counts (b_{01}, b_{10}) and two-sided exact McNemar p-values.

Table S3. Complete inventory of the 18 configurable GSPA v2.1.0 hyperparameters across Stages 1 and 2B.

Stage / Module	Symbol	Parameter Key	Source Module	Role in Inference	Default Value
Stage 1 (B_1)	α_{ess}	`alpha_ess`	`EssentialityPrior.groovy`	Unfulfilled essential GO function log-odds boost	1.50
Stage 1 (B_2)	α_{coh}	`alpha_coh`	`CoherencePrior.groovy`	Active pathway coherence log-odds scale	1.00
Stage 1 (B_2)	w_{part}	`alpha_complex_boost`	`CoherencePrior.groovy`	Obligate heteromeric complex partner boost ($0.65\,w_{\text{part}} = 0.5525$ for $N_{\text{strong}} \geq 2$)	0.85
Stage 1 (B_2)	w_{sing}	`alpha_complex_penalty`	`CoherencePrior.groovy`	Isolated obligate heteromeric singleton penalty magnitude (applied as -0.75)	0.75
Stage 1 (B_3)	α_{cons}	`alpha_cons`	`ConsistencyPrior.groovy`	SAT4J taxonomic conflict penalty magnitude (applied as -3.00)	3.00
Stage 1 (B_4)	α_{anat}	`alpha_anat_penalty`	`AnatomicalPrior.groovy`	Taxon-forbidden Uberon/PO/CL structure penalty magnitude (applied as -3.50)	3.50
Stage 1 (B_4)	α_{boost}	`alpha_anat_boost`	`AnatomicalPrior.groovy`	Active-structure developmental rescue log-odds boost	0.65
Stage 1 (B_5)	α_{gap}	`alpha_gap`	`GapFillingPrior.groovy`	Metabolic pathway hole-filling log-odds boost	1.50
Stage 1 (B_6)	α_{ctx}	`alpha_ctx`	`GenomicContextPrior.groovy`	Co-operonic pathway enrichment log-odds scale	1.00
Stage 1 (B_6)	$\alpha_{\text{gap_ctx}}$	`alpha_gap_ctx`	`GenomicContextPrior.groovy`	Co-operonic metabolic gap multiplier on $\alpha_{\text{ctx}}\, c_O(W)$	1.50
Stage 1 (Iter)	γ_{iter}	`damping`	`IterativeRefiner.groovy`	Iterative log-odds fixed-point damping weight	0.50
Stage 1 (Iter)	$K_{\max}$	`max_iter`	`IterativeRefiner.groovy`	Maximum Stage 1 message-passing iterations	6
Stage 1 (Iter)	ϵ	`epsilon`	`IterativeRefiner.groovy`	Stage 1 probability convergence tolerance	0.005
Stage 2B (SLR)	τ_{core}	`tau_core`	`stoichiometric_locus_refiner.py`	Core reaction probability threshold $S_{\text{rxn}}(j)$	0.50
Stage 2B (SLR)	α_{bio}	`alpha_bio`	`stoichiometric_locus_refiner.py`	Minimum required fraction of $v_{\text{bio}}^{\max}$ in pFBA	0.10
Stage 2B (SLR)	η_{null}	`tau_null`	`stoichiometric_locus_refiner.py`	Unassigned-gene hypothesis $h_{\varnothing}$ prior mass	0.05
Stage 2B (SLR)	γ_{FBA}	`gamma_fba`	`stoichiometric_locus_refiner.py`	Backward locus responsibility log-odds scale	2.20
Stage 2B (SLR)	γ_{damp}	`gamma_damp`	`stoichiometric_locus_refiner.py`	Geometric damping decay across outer passes	0.60

Table S4. CAFA temporal holdout evaluation across 16,125 protein–aspect evaluations (3,440 twilight-zone evaluations at $< 30\%$ sequence identity).

Method	MF All	MF $< 30\%$	BP All	BP $< 30\%$	CC All	CC $< 30\%$	Macro $F_{\max}$ All	Macro $F_{\max} < 30\%$	Runtime / Peak RAM
DIAMOND (BLAST)	0.6224	0.0000	0.4437	0.0000	0.5805	0.0000	0.5489	0.0000	4.2 s / 1,150 MB
DeepGOCNN	0.4280	0.4209	0.3431	0.3389	0.6409	0.6487	0.4707	0.4695	8.5 s / 2,400 MB
DeepGOPlus (CNN+BLAST)	0.6337	0.4207	0.4613	0.3364	0.6722	0.6468	0.5891	0.4680	12.7 s / 2,650 MB
DeepGOZero	0.6552	0.5006	0.4514	0.3614	0.6602	0.6327	0.5889	0.4982	9.4 s / 2,800 MB
InterPro MLP	0.6561	0.5066	0.4592	0.3707	0.6667	0.6384	0.5940	0.5052	14.2 s / 3,100 MB
DeepGOZero+BLAST	0.6676	0.4982	0.4816	0.3614	0.6668	0.6327	0.6053	0.4974	13.6 s / 2,950 MB
ProteInfer	0.6966	0.6810	0.6146	0.5948	0.5301	0.5181	0.6138	0.5980	38.4 s / 6,400 MB
Stage 0 Noisy-OR (Ensemble)	0.7159	0.6949	0.6387	0.6144	0.7241	0.7075	0.6929	0.6723	56.8 s / 6,400 MB
GSPA (Iter 1)	0.7163	0.6971	0.6391	0.6152	0.7257	0.7202	0.6937	0.6775	59.2 s / 6,450 MB
GSPA (Iter 2)	0.7164	0.6972	0.6365	0.6143	0.7259	0.7202	0.6929	0.6772	61.5 s / 6,450 MB
GSPA (Iter 3)	0.7165	0.6972	0.6332	0.6104	0.7259	0.7202	0.6919	0.6759	63.8 s / 6,450 MB

Table S5. Complete post-cutoff Swiss-Prot enzyme evaluation (Holdout-125, $n = 125$ proteins, 293 ground-truth EC annotations).

Predictor / Refinement Method	Masking Tier	Top-1 Acc. (Hits)	Top-3 Acc. (Hits)	Top-5 Acc. (Hits)	MRR	Micro $F_{\max}$
CLEAN (2023)	Unfiltered	32.8% (41/125)	40.8% (51/125)	41.6% (52/125)	0.3719	0.4360
DeepProZyme (2023)	Unfiltered	23.2% (29/125)	31.2% (39/125)	32.0% (40/125)	0.2750	0.2715
DeepProZyme + METEOR v8 (Single-Pred)	Unfiltered	25.6% (32/125)	31.2% (39/125)	31.2% (39/125)	0.2872	0.2804
DeepProZyme + GSPA Single-Pred (SLR + Coh.)	Unfiltered	23.2% (29/125)	31.2% (39/125)	32.8% (41/125)	0.2779	0.2703
EnzBERT (2023)	Unfiltered	32.8% (41/125)	37.6% (47/125)	39.2% (49/125)	0.3531	0.3596
Noisy-OR (3-Method Ensemble)	Unfiltered	33.6% (42/125)	40.8% (51/125)	43.2% (54/125)	0.3807	0.4381
GSPA (3-Method, Iter 1)	Unfiltered	32.0% (40/125)	42.4% (53/125)	44.0% (55/125)	0.3775	0.4035
GSPA (3-Method, Iter 2)	Unfiltered	32.0% (40/125)	42.4% (53/125)	44.0% (55/125)	0.3775	0.3891
GSPA (3-Method, Iter 3)	Unfiltered	32.0% (40/125)	43.2% (54/125)	44.0% (55/125)	0.3781	0.3830
EnzBERT	$\leq 70\%$ identity	30.4% (38/125)	35.2% (44/125)	35.2% (44/125)	0.3305	0.3204
EnzBERT	$\leq 50\%$ identity	28.0% (35/125)	33.6% (42/125)	33.6% (42/125)	0.3040	0.3163
EnzBERT	$\leq 30\%$ identity	17.6% (22/125)	24.8% (31/125)	24.8% (31/125)	0.2139	0.2212
CLEAN	$\leq 30\%$ identity	24.0% (30/125)	30.4% (38/125)	32.8% (41/125)	0.2741	0.2299
DeepProZyme	$\leq 70\%$ identity	23.2% (29/125)	31.2% (39/125)	32.0% (40/125)	0.2713	0.2715
DeepProZyme	$\leq 50\%$ identity	23.2% (29/125)	31.2% (39/125)	32.8% (41/125)	0.2720	0.2684
DeepProZyme	$\leq 30\%$ identity	22.4% (28/125)	31.2% (39/125)	32.8% (41/125)	0.2674	0.2650

Table S6. Single-predictor and 3-channel consensus acceptance filtering of candidate METEOR v8 EC changes across 95 bacterial genomes under four donor-database homology-masking tiers. p-values report two-sided exact binomial McNemar tests on discordant Top-1 predictions (b_{01}/b_{10}).

Predictor	Tier	Genomes / Proteins	Base Top-1 ($F_{\max}$)	METEOR v8 Top-1 (Δ)	METEOR b_{01}/b_{10} (p)	GSPA Single-Pred Top-1 (Δ)	Single-Pred b_{01}/b_{10} (p)	GSPA Multi-Pred Top-1 (Δ)	Multi-Pred b_{01}/b_{10} ($F_{\max}$)
CLEAN	Unfiltered	73/52,814	0.8214 (0.8796)	0.8234 (+108)	184/76 ($< 10^{-6}$)	0.8228 (+77)	99/22 ($< 10^{-6}$)	0.8243 (+155)	162/7 (0.8816)
CLEAN	$\leq 70\%$	95/69,663	0.8267 (0.8836)	0.8289 (+149)	191/42 ($< 10^{-6}$)	0.8283 (+108)	131/23 ($< 10^{-6}$)	0.8289 (+149)	154/5 (0.8851)
CLEAN	$\leq 50\%$	95/69,663	0.8271 (0.8844)	0.8294 (+162)	208/46 ($< 10^{-6}$)	0.8291 (+139)	156/17 ($< 10^{-6}$)	0.8293 (+155)	158/3 (0.8859)
CLEAN	$\leq 30\%$	94/68,681	0.8288 (0.8833)	0.8306 (+122)	156/34 ($< 10^{-6}$)	0.8302 (+93)	107/14 ($< 10^{-6}$)	0.8306 (+126)	132/6 (0.8845)
DeepProZyme	Unfiltered	95/69,666	0.9022 (0.9331)	0.9010 (−85)	103/188 (1.0×10^{-6})	0.9017 (−33)	24/57 (3.2×10^{-4})	0.9027 (+36)	51/15 (0.9335)
DeepProZyme	$\leq 70\%$	95/69,666	0.8965 (0.9290)	0.8961 (−27)	151/178 (0.152)	0.8963 (−15)	44/59 (0.167)	0.8975 (+67)	78/11 (0.9296)
DeepProZyme	$\leq 50\%$	95/69,666	0.8966 (0.9288)	0.8966 (+1)	163/162 (1.000)	0.8966 (−3)	49/52 (0.842)	0.8979 (+87)	93/6 (0.9297)
DeepProZyme	$\leq 30\%$	95/69,666	0.8967 (0.9289)	0.8970 (+19)	170/151 (0.315)	0.8966 (−4)	51/55 (0.771)	0.8976 (+67)	76/9 (0.9296)
EnzBERT	Unfiltered	95/69,666	0.8430 (0.8707)	0.8416 (−92)	100/192 ($< 10^{-6}$)	0.8428 (−10)	51/61 (0.395)	0.8438 (+59)	65/6 (0.8713)
EnzBERT	$\leq 70\%$	95/69,666	0.8394 (0.8675)	0.8374 (−146)	96/242 ($< 10^{-6}$)	0.8391 (−26)	41/67 (0.0157)	0.8405 (+70)	76/6 (0.8682)
EnzBERT	$\leq 50\%$	95/69,666	0.8330 (0.8602)	0.8304 (−181)	132/313 ($< 10^{-6}$)	0.8324 (−38)	31/69 (1.8×10^{-4})	0.8343 (+89)	93/4 (0.8611)
EnzBERT	$\leq 30\%$	95/69,666	0.8256 (0.8515)	0.8232 (−167)	108/275 ($< 10^{-6}$)	0.8254 (−18)	39/57 (0.082)	0.8267 (+78)	80/2 (0.8524)

S3 Cross-ontology bridge and multi-kingdom population evaluations

Supplementary Table S7 summarizes the five resource tables programmatically extracted from the official OBO Foundry releases of the Gene Ontology (GO), Uberon multi-species anatomy ontology, Plant Ontology (PO), Cell Ontology (CL), and NCBI Taxonomy and loaded at runtime by the taxonomic and anatomical constraint modules. Supplementary Table S8 reports the 1,000-genome RefSeq bacterial population audit, and Supplementary Table S9 reports the 88-proteome multi-kingdom evaluation across Archaea, Bacteria, Eukaryota, and Viruses.

Table S7. Summary of programmatic cross-ontology constraint tables in GSPA v2.1.0.

Resource File	Ontologies Bridged	Data Rows	Distinct Classes	Constraint Semantics & Examples
`go-taxon-constraints.tsv`	GO ↔ NCBITaxon	26,911	18,159 GO terms	13,819 `only_in_taxon` + 13,092 `never_in_taxon` rules (e.g., `GO:0009579` thylakoid)
`go-anatomical-constraints.tsv`	GO ↔ Uberon / PO / CL	2,715	2,674 GO; 1,294 Anat.	30 object properties (`results_in_development_of`, `occurs_in`, etc.)
`uberon-po-taxon-constraints.tsv`	Uberon / PO / CL ↔ NCBITaxon	8,331	1,247 Anat. classes	4,767 `only_in_taxon` + 3,564 `never_in_taxon` rules (e.g., `UBERON:0000948` heart → Metazoa)
`uberon-po-hierarchy.tsv`	Uberon / PO / CL DAG	7,722	2,855 Anat. classes	Transitive `is_a` and `part_of` edges across anatomical structures
`ncbi-taxon-hierarchy.tsv`	NCBI Taxonomy DAG	1,818	1,305 Taxon clades	Transitive taxonomic `is_a` lineage closure for benchmark clades

Table S8. Population-scale structural audit across 1,000 RefSeq bacterial genomes comparing deep-learning GO predictors, InterProScan, Stage 0 pooling, and full GSPA.

Method	Genomes (n)	Macro $F_{\max}$	Essential Core-22 (%)	Essential Full-29 (%)	Complex Coh. (%)	MetaCyc Comp. (%)	Taxon Viol. / Genome	Runtime / Peak RAM
DeepGO-SE / DeepGOMeta (2024)	1,000	0.5047	95.34%	82.17%	79.83%	11.59%	14,198.8	14.2 s / 4,200 MB
SPROF-GO (2023)	1,000	0.4930	99.91%	83.70%	79.18%	0.18%	7,322.7	28.5 s / 6,100 MB
DeepGraphGO (2021)	1,000	0.4020	99.28%	85.72%	70.56%	37.37%	9,922.2	19.4 s / 5,400 MB
DeepFRI (2021)	1,000	0.3843	100.00%	81.25%	98.87%	9.43%	106,815.2	24.0 s / 4,800 MB
TALE (2021)	1,000	0.3883	87.24%	75.38%	65.73%	19.63%	9,794.7	16.8 s / 4,500 MB
InterProScan 5	1,000	0.5940	97.36%	86.90%	61.89%	40.35%	940.4	145.0 s / 8,200 MB
GSPA Stage 0 (Noisy-OR)	1,000	0.6929	99.81%	90.42%	66.03%	36.71%	23,384.1	0.85 s / 3,400 MB
GSPA Stage 1+2 (Full)	1,000	0.6937	99.99%	95.29%	100.00%	66.75%	0.0	5.4 s / 4,812 MB

Table S9. Multi-kingdom proteome-wide evaluation across 88 proteomes (121,631 proteins) spanning Archaea, Bacteria, Eukaryota, and Viruses under homology masking.

Domain	Configuration	Genomes / Proteins	Mean $F_{\max}$ (All)	Mean $F_{\max}$ (< 30%)	MF / BP / CC $F_{\max}$	Coverage	$\Delta F_{\max}$ [95% Bootstrap CI]	Runtime / Heap	Stage 3 Trigger (%)
Archaea	Stage 0 Noisy-OR	15/8,639	0.9125	0.5419	0.9460/0.9341/0.9146	0.9524	Baseline	0.35 s / 4,746 MB	1.97%
	GSPA Stage 1+2A	15/8,639	0.9129	0.5422	0.9454/**0.9345**/**0.9163**	0.9524	+0.0004 [−0.0001, +0.0010]	2.29 s / 5,120 MB	1.97%
Bacteria	Stage 0 Noisy-OR	47/104,032	0.7987	0.7987	0.8326/0.8304/0.8322	0.8443	Baseline	2.16 s / 4,479 MB	1.72%
	GSPA Stage 1+2A	47/104,032	0.8009	0.8009	0.8319/**0.8316**/**0.8371**	0.8443	+0.0022 [+0.0014, +0.0031]	12.66 s / 4,812 MB	1.72%
Eukaryota	Stage 0 Noisy-OR	6/6,155	0.7096	0.2459	0.8382/0.6709/0.7520	0.9079	Baseline	1.13 s / 5,676 MB	2.23%
	GSPA Stage 1+2A	6/6,155	0.7117	0.2453	0.8380/**0.6732**/**0.7555**	0.9079	+0.0021 [−0.0006, +0.0085]	6.67 s / 4,232 MB	2.23%
Viruses	Stage 0 Noisy-OR	20/2,805	0.7222	0.3112	0.8303/0.7726/0.7004	0.7576	Baseline	0.27 s / 4,841 MB	1.90%
	GSPA Stage 1+2A	20/2,805	0.7252	0.3116	**0.8304**/**0.7747**/**0.7037**	0.7576	+0.0030 [+0.0004, +0.0067]	0.42 s / 4,575 MB	1.90%

S4 Secondary metabolism, metagenomic dark matter, and Stage 2B stoichiometric refinement

In the Price-149 enzyme evaluation, the cognate-substrate stoichiometric neighborhood probe and outer-loop damped anchor are defined as follows. For each candidate pair (p, u), let $n_1(p, u)$ and $n_2(p, u)$ count chromosomal neighbors within ± 10 genes whose reactions share non-currency metabolites with u at 1 and 2 hops, $f_{\mathrm{op}}(p)$ denote the co-directional operon fraction, and $s_{\mathrm{nb}}(p, u) = \min(2.0, \sum_{|g'-g|\le 10}(1 + 0.25|g' - g|)^{-1} P_{\mathrm{cand}}(g', u))$ measure distance-weighted neighbor EC support. The local stoichiometric probe shift $\Delta L_{\mathrm{probe}}(p, u) = 0.85 \min(1.5, 0.55n_1 + 0.20n_2) + 0.45 f_{\mathrm{op}}(p)\mathbb{I}[n_1 + n_2 > 0] + 0.35\mathbb{I}[u \in \mathcal{R}_{\mathrm{bal}}] - 0.50\mathbb{I}[u \notin \mathcal{R}_{\mathrm{ec}}]$ and Stage 1 neighbor coherence shift $\Delta L_{\mathrm{stage1}}(p, u) = 0.90 s_{\mathrm{nb}}(p, u) + 0.50 f_{\mathrm{op}}(p)\mathbb{I}[s_{\mathrm{nb}}(p, u) > 0.20] + 0.30\mathbb{I}[u \in \mathcal{W}_{\mathrm{kegg}}]$ are applied over $T = 3$ damped outer iterations ($\lambda_t \in \{0.70, 0.42, 0.252\}$):

$$P^{(t)}_{\mathrm{single}}(p, u) = \sigma_{\log}\Big(0.75 \operatorname{logit}\big(P^{(t-1)}(p, u)\big) + 0.25 \operatorname{logit}\big(P^{(0)}(p, u)\big) + \lambda_t\big(\Delta L_{\mathrm{stage1}}(p, u) + 0.65\, \Delta L_{\mathrm{probe}}(p, u)\big)\Big), \quad \text{(S1)}$$

and the multi-predictor posterior blends $0.55 P^{(t)}_{\mathrm{single}}(p, u) + 0.45 P_{\mathrm{NoisyOR}}(p, u)$. Supplementary Table S10 reports the biosynthetic gene cluster benchmark across 106 MIBiG/antiSMASH clusters, Supplementary Table S11 reports the 97-MAG uncultured microbial dark-matter panel across 16 phyla, and Supplementary Table S12 reports the 108-genome stoichiometric invariant audit, the comparison across six manually curated BiGG reference genome-scale metabolic models, and the role-stratified Price-149 breakdown on CLEAN.

Table S10. Biosynthetic gene cluster (BGC) recovery and non-cluster chromosomal specificity across 106 curated clusters from 12 bacterial species.

BGC Class	Clusters (n)	Core / Tailoring Loci	Stage 0 Core Rec.	GSPA Core Rec.	Stage 0 Tailoring Rec.	GSPA Tailoring Rec.	Non-BGC Specificity / Overall F_1 (Stage 0 → GSPA)
NRPS (Non-ribosomal peptide synthetase)	38	114/266	0.7807	0.9211	0.5414	0.8383	99.58% / 0.7420 → **0.8962**
Type I / II PKS (Polyketide synthase)	29	87/203	0.7586	0.9080	0.5222	0.8227	99.62% / 0.7227 → **0.8865**
Terpene / Carotenoid	18	36/90	0.8056	0.9444	0.5667	0.8556	99.69% / 0.7619 → **0.9136**
Siderophore (Aerobactin-like)	12	36/72	0.8333	0.9722	0.5972	0.8889	99.72% / 0.7935 → **0.9384**
RiPP / Bacteriocin / Lantipeptide	9	18/54	0.7222	0.8889	0.4815	0.7963	99.55% / 0.6842 → **0.8613**
All 106 BGCs	106	291/685	0.7801	0.9244	0.5387	0.8380	99.62% / 0.7395 → 0.8959

Table S11. Pathway gap resolution across 97 high-quality metagenome-assembled genomes (MAGs) of uncultured bacteria and archaea spanning 16 phyla.

Phylum / Group	Domain	MAGs (n)	Candidate Fills ($q \geq 0.30$)	High-Conf. Fills ($q \geq 0.50$)	Mean Fills / MAG ($q \geq 0.30/0.50$)	Pathway Comp. Before	Pathway Comp. After
Pseudomonadota	Bacteria	19	502	137	26.4/7.2	0.542	0.684
Actinomycetota	Bacteria	12	298	82	24.8/6.8	0.518	0.661
Bacteroidota	Bacteria	11	237	61	21.5/5.5	0.491	0.642
Bacillota	Bacteria	9	208	58	23.1/6.4	0.528	0.669
Cyanobacteriota	Bacteria	8	158	39	19.8/4.9	0.534	0.675
Chloroflexota	Bacteria	7	143	36	20.4/5.1	0.462	0.618
Halobacteriota	Archaea	7	132	32	18.9/4.6	0.489	0.631
Thermoproteota	Archaea	6	105	25	17.5/4.2	0.495	0.638
Other 8 Phyla (CPR, Asgard, etc.)	Mixed	18	392	98	21.8/5.4	0.469	0.613
All 97 Uncultured MAGs	**All (16 Phyla)**	97	2,175	568	22.4/5.9	0.511	0.656

Table S12. 108-genome stoichiometric invariant audit, curated BiGG reference GEM comparison ($n = 6$ species), and role-stratified Price-149 CLEAN refinement across Strata A, B, and C.

Panel / Stratum	Method / Reconstruction Pipeline	Active / Growing	Added Rxns (Cert. / Pad.)	Imbal. / Dead-End Added	EC Prec. / Rec. / F_1	BiGG Rxn F_1	Top-1 Hits (Acc.)
108-Genome Audit	Raw Sequence Threshold (≥ 0.50)	2,809.5 (0% grow)	0.0 (0.0/0.0)	2,621.7 active / 9.75	n/a	n/a	n/a
	METEOR v8 (Negative-Cost MILP)	3,121.0 (100% grow)	313.35 (51.01/262.34)	2,597.8 active / 2.36	n/a	n/a	n/a
	Grow-Support Threshold Repair (Pos.-Cost LP)	2,584.8 (100% grow)	81.31 (81.31/0.00)	0.0 added / 2.23	n/a	n/a	n/a
	GSPA Stage 2B (Lexicographic pFBA)	2,815.2 (300.6 **flux**, 100% **grow**)	5.70 (5.70/0.00)	0.0 **added** / 0.00	n/a	n/a	n/a
6 Curated BiGG GEMs	CarveMe v1.5.2 (Native BiGG Database)	6/6 (100%)	n/a	n/a	0.8217/0.8927/0.8536	0.6955	n/a
	Reconstructor (ModelSEED Namespace)	4/6 (67%)	n/a	n/a	0.4387/0.8787/0.5844	n/a	n/a
	DeepProZyme + METEOR v8 (Neg.-Cost MILP)	6/6 (100%)	n/a	n/a	0.4383/0.8842/0.5851	0.2091	n/a
Price-149 Stratum A ($n = 91$)	CLEAN Base → + Stage 2B Probe → + Single-Pred → + **Multi-Pred**	Core Biomass	n/a	n/a	$F_{\max}$: 0.6336 → 0.6405 → 0.6274 → **0.7019**	Macro: 0.4740 → **0.5288**	46 → 46 → 47 → **52** (57.1%)
Price-149 Stratum B ($n = 55$)	CLEAN Base → + Stage 2B Probe → + Single-Pred → + **Multi-Pred**	Catabolic	n/a	n/a	$F_{\max}$: 0.3580 → 0.3753 → 0.3461 → **0.4546**	Macro: 0.3103 → **0.3436**	13 → 13 → **15** (27.3%) → 13 (23.6%)
Price-149 Stratum C ($n = 2$)	CLEAN Base → + Stage 2B Probe → + Single-Pred → + **Multi-Pred**	Secondary	n/a	n/a	$F_{\max}$: 0.5099 → 0.5099 → 0.5175 → **0.5833**	Macro: 0.5000 → 0.5000	1 → 1 → 1 → 1 (50.0%)

S5 Experimental phenotype prediction in BacDive and statistical testing

In the BacDive benchmark, the five-fold genus-blocked cross-validation assigns each taxonomic genus deterministically to one fold via MD5 hashing. Within each training fold, tool-specific conditional probabilities $p_{g,\mathrm{cal}}, p_{m,\mathrm{cal}}$, assay prevalence $a_{\mathrm{prev}}(y)$, and fold-local pooled probability $p_{\mathrm{pool}}(s, y)$ ($0.38p_{g,\mathrm{cal}} + 0.18p_{m,\mathrm{cal}} + 0.22a_{\mathrm{prev}} + 0.22P_{\mathrm{Track1}}$ for carbon; $0.36p_{g,\mathrm{cal}} + 0.16p_{m,\mathrm{cal}} + 0.22a_{\mathrm{prev}} + 0.26P_{\mathrm{Track1}}$ for enzymes) are estimated strictly from the training split. Supplementary Table S13 reports the per-fold five-fold genus-blocked cross-validation metrics across both panels.

Table S13. Summary of 5-fold genus-blocked cross-validation in BacDive across carbon-source growth ($n = 2{,}332$ assays, 529 strains) and diagnostic enzymes ($n = 17{,}496$ assays, 3,017 type strains).

Assay Panel	Classifier	Fold 0 F_1	Fold 1 F_1	Fold 2 F_1	Fold 3 F_1	Fold 4 F_1	5-Fold Mean ± s.d.	Pooled F_1 (BAcc / ROC-AUC)
Carbon Growth ($n = 2{,}332$)	BacDive-AI RF Baseline	0.8709	0.8720	0.8397	0.7174	0.8646	0.8329 ± 0.0659	0.8448 (0.8842/0.9462)
	DiASPora GBDT	0.8582	0.8819	0.8582	0.7817	0.8238	0.8408 ± 0.0390	0.8483 (0.8879/0.9519)
	Traitar L1-SVM	0.8480	0.8765	0.8727	0.7708	0.8653	0.8467 ± 0.0438	0.8571 (0.8888/0.9430)
	GSPA + DiASPora Hybrid	0.8830	0.9048	0.8824	0.7778	0.8602	0.8616 ± 0.0495	0.8710 (0.9035/0.9567)
Diagnostic Enzymes ($n = 17{,}496$)	BacDive-AI RF Baseline	0.8477	0.8371	0.8490	0.8553	0.8387	0.8456 ± 0.0076	0.8458 (0.8427/0.8962)
	DiASPora GBDT	0.8465	0.8579	0.8560	0.8655	0.8493	0.8550 ± 0.0075	0.8550 (0.8535/0.9061)
	Traitar L1-SVM	0.8620	0.8561	0.8562	0.8714	0.8517	0.8595 ± 0.0076	0.8593 (0.8552/0.9034)
	GSPA + DiASPora Hybrid	0.8671	0.8663	0.8718	0.8854	0.8653	0.8712 ± 0.0083	0.8713 (0.8714/0.9097)

S6 Stage 3 neuro-symbolic tool schemas, multi-seed stability, prospective CAFA6, and extremophile case study

Stage 3 provides five deterministic, read-only genomic and ontological inspection tools to the open-weight instruction model (`run_gspa_agent.py`):

1. `inspect_operon_neighborhood(protein_id, window=5)`: Returns the de-identified genomic neighborhood (±5 genes on the same contig), intergenic distances, strand orientation, and neighbor GO/EC annotations.

2. `inspect_pathway_or_complex_gap(anomaly_id)`: Returns the target pathway or heteromeric complex identifier, fraction of partner steps or subunits present in the genome at $P \geq 0.50$, and list of missing steps or

subunits.

3. `query_protein_evidence(protein_id)`: Returns Stage 0 per-channel calibrated probabilities, Pfam/InterPro structural domain annotations, and precomputed RLGC kernel density Bayes factors ($\mathrm{BF}_{\mathrm{operon}}$).

4. `check_symbolic_constraints(go_or_ec_term, taxon_id)`: Queries the deterministic ontology bridge (26,911 direct GO taxon rules and 2,715 GO–Uberon/PO/CL anatomical rules linked to 8,331 taxon constraints across 7,722 anatomical and 1,818 taxonomic hierarchy edges) and returns whether the term is valid in `taxon_id`, along with its nearest valid non-root GO ancestors if forbidden.

5. `propose_resolution(action, target_protein_id, resolved_term, confidence, justification)`: Submits the final structured decision (PROMOTE, RELAX, DEMOTE, or ABSTAIN), which is deterministically verified by the symbolic guardrail before acceptance.

Supplementary Table S14 reports the per-seed and three-seed summary of the full guarded multi-turn Stage 3 configuration across seeds 42, 43, and 44 on the 120-anomaly and 50-unique-protein LRO benchmarks, Supplementary Table S15 reports the prospective CAFA6 evaluation across 769 targets, and Supplementary Table S16 summarizes the five desert extremophile isolates from the Empty Quarter (*Rub' al Khali*).

Table S14. Three-seed evaluation of Stage 3 guarded multi-turn reasoning across the 120-anomaly and 50-gap LRO benchmarks.

Evaluation Run	Random Seed	Precision	Recall	F_1 Score	Accuracy	Violations Accepted	LRO-50 Hit@1 (MRR)
Stage 3 Multi-Turn + Guardrail (120 Anomalies + LRO-50)	Seed 42	0.9787	0.8519	0.9109	0.9250	0/120 (0.0%)	82.00% (0.8740)
Stage 3 Multi-Turn + Guardrail (120 Anomalies + LRO-50)	Seed 43	0.9592	0.8704	0.9126	0.9333	0/120 (0.0%)	84.00% (0.8890)
Stage 3 Multi-Turn + Guardrail (120 Anomalies + LRO-50)	Seed 44	0.9787	0.8519	0.9109	0.9250	0/120 (0.0%)	82.00% (0.8747)
Stage 3 Multi-Turn + Guardrail (3-Seed Mean ± s.d.)	**Seeds 42, 43, 44**	0.9722 ± 0.0092	0.8581 ± 0.0087	0.9115 ± 0.0008	0.9278 ± 0.0039	$0.00 \pm 0.00\%$	$82.67 \pm 0.94\%$ (0.8792 ± 0.0069)

Table S15. Prospective CAFA6 evaluation across 769 targets (412 no-knowledge and 357 limited-knowledge targets).

Method	Targets (n)	MF $F_{\max}$ ($\mathrm{WF}_{\max}$)	BP $F_{\max}$ ($\mathrm{WF}_{\max}$)	CC $F_{\max}$ ($\mathrm{WF}_{\max}$)	Macro $F_{\max}$	Macro $\mathrm{WF}_{\max}$	NK Macro $F_{\max}$	LK Macro $F_{\max}$	Runtime / Peak RAM
ESM-2 + ProstT5 3Di MLP (2025)	769	0.4520 (0.3840)	0.3700 (0.3420)	0.6650 (0.5860)	0.4957	0.4370	0.4612	0.5355	18.4 s / 5,200 MB
Agentic AFP + Domain/EC (Qwen3-14B, 2026)	769	0.4810 (0.4100)	0.3990 (0.3700)	0.6590 (0.5590)	0.5130	0.4463	0.4795	0.5516	24.0 s / 6,800 MB
GSPA Stage 3 Neuro-Symbolic Agent	769	0.5470 (0.4980)	0.4140 (0.3720)	0.6690 (0.5820)	0.5433	0.4840	0.5124	0.5790	19.8 s / 5,400 MB

Table S16. Empty Quarter (*Rub' al Khali*) desert extremophile isolate annotation summary across five bacterial genomes.

Empty Quarter Isolate	Taxonomic Clade	Proteins (N)	Raw C_{ess}	Stage 0 C_{ess}	GSPA C_{ess}	Path. Coh. (Raw → GSPA)	pFBA Repairs / Stage 3 Prom.
Salinicola sp. EQ-01	Gammaproteobacteria	3,842	0.8810	0.9120	0.9680	0.4820 → 0.6140	6 repairs / 5 promotions
Halomonas sp. EQ-02	Gammaproteobacteria	4,118	0.8920	0.9210	0.9740	0.4950 → 0.6280	5 repairs / 6 promotions
Pseudomonas sp. EQ-03	Gammaproteobacteria	5,490	0.9140	0.9380	0.9820	0.5190 → 0.6410	4 repairs / 4 promotions
Bacillus endophyticus EQ-04	Bacillota	4,625	0.9010	0.9280	0.9760	0.5040 → 0.6320	5 repairs / 7 promotions
Streptomyces sp. EQ-05	Actinomycetota	7,184	0.8540	0.8960	0.9590	0.4460 → 0.5980	8 repairs / 9 promotions
5-Isolate Mean	**3 Phyla / Classes**	5,051.8	0.8884	0.9190	0.9718	0.4892 → 0.6226	5.6 **repairs** / 6.2 **promotions**